

Question 1: Supermarket Customer Segmentation & Targeted Marketing

Dataset: Supermarket Purchases

Customer	Visits / Month	Avg Spend (₹)	Items Bought
C1	10	2,000	15
C2	4	5,000	8
C3	12	1,800	18
C4	3	6,000	6
C5	11	2,100	16

1. Proposed Clustering Approach: K-Means Clustering

K-Means Clustering is recommended for segmenting this supermarket dataset due to its efficiency with continuous numeric attributes and spherical cluster geometry.

• Step 1: Feature Standardization (Z-Score Normalization):

The features have vastly different scales (Avg Spend ranges in thousands, while Visits and Items are small counts). We apply standard scaling:

$z = (x - \mu) / \sigma$ to ensure distance metrics (Euclidean distance) are not dominated by Avg Spend.

• Step 2: Optimal Cluster Determination (k = 2):

Using the Elbow Method and Silhouette Analysis, the optimal number of clusters is identified as k = 2.

• Step 3: Cluster Formation & Characteristics:

- Cluster 1 (Frequent Grocery Shoppers - C1, C3, C5): Characterized by High Visits (~11/month), High Items Bought (~16 items), and Moderate/Budget Spend (~₹1,967).
- Cluster 2 (High-Value Bulk/Premium Shoppers - C2, C4): Characterized by Low Visits (~3.5/month), Low Item Count (~7 items), and High Spend (~₹5,500).

2. Targeted Marketing Strategies Based on Segmentation

Segment / Cluster	Customer Profile	Targeted Marketing Actionable Strategy
Cluster 1 (C1, C3, C5)	High Frequency, Budget / Routine Shoppers	• Loyalty & Basket-Size Expansion: Offer tiered bundle discounts (e.g., 'Spend ₹2,500 and get ₹300 off') to increase basket spend. • Frequency Rewards: Digital stamp cards or weekly routine coupons on household essentials and daily perishables.

Cluster 2 (C2, C4)	Low Frequency, High-Value Premium Shoppers	• Premium Engagement: VIP loyalty perks, personalized concierge alerts for premium brand arrivals, gourmet foods, and imported goods. • Win-Back & Frequency Stimulation: Fortnightly SMS/App reminders with exclusive discounts to encourage an additional monthly visit.
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Question 2: Fraud & Anomaly Detection in Credit Card Transactions

Dataset: Credit Card Transactions

Transaction	Amount (₹)	Location	Time (hrs)
T1	200	Chennai	10:00
T2	5,000	Delhi	02:00
T3	150	Chennai	11:00
T4	7,000	Mumbai	01:00
T5	180	Chennai	12:00

1. Proposed Clustering Solution: DBSCAN / Isolation Forest

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is the most suitable clustering algorithm for fraud detection because:

- 1. It does not force all data points into clusters; it naturally labels low-density points as Noise / Outliers (-1).
- 2. It effectively separates dense routine clusters from rare, suspicious activities.

2. Analytical Data Preprocessing & Distance Formulation

- Feature Engineering: Location is encoded (e.g., Target Frequency Encoding or Geospatial Coordinates from primary residence). Time is treated cyclically using sine/cosine transformations.
- Feature Scaling: Features are normalized using RobustScaler to prevent extreme fraudulent outliers from skewing mean and variance.

3. Anomaly Analysis & Detection Results

- Normal Transaction Cluster (T1, T3, T5):
 - Profile: Small transaction amounts (₹150 – ₹200), domestic primary location (Chennai), standard daytime business hours (10:00 – 12:00).
 - Classification: Dense Core Cluster (Legitimate User Activity).
- Anomaly / Fraud Outliers (T2, T4):

- T2 (Delhi, 02:00 hrs, ₹5,000): Anomaly due to unusual late-night timing, geographic deviation from Chennai baseline, and 25x higher spend.
- T4 (Mumbai, 01:00 hrs, ₹7,000): Severe anomaly with high financial risk, out-of-city location, and midnight execution time.
- Classification: DBSCAN Noise / Outlier Points.

4. Real-Time Actionable Fraud Mitigation Framework

1. Instant Friction / Step-Up Authentication: Automatically trigger Multi-Factor Authentication (OTP, biometric) or temporary authorization holds on transactions falling into noise clusters.
2. Velocity & Impossible Travel Check: Flag physical distance discrepancies between consecutive transactions (e.g., Chennai at 12:00 vs Mumbai at 01:00).
3. Risk Scoring Engine: Assign higher fraud risk scores to transactions with high distance from established user cluster centroids.

Question 3: Telecom Churn Prediction & Retention Strategies

Dataset: Telecom Usage

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5.0	200
U2	100	2.0	100
U3	350	6.0	220
U4	80	1.0	90
U5	320	5.5	210

1. Clustering Methodology: Agglomerative Hierarchical / K-Means Clustering

After normalizing Call Duration, Data Usage, and Recharge Amount using Min-Max Scaling, clustering groups users into distinct behavioral cohorts:

Cluster Segment	Users	Usage Profile	Churn Risk Level
Active / Engaged Cohort	U1, U3, U5	High Calls (300-350 min), High Data (5-6 GB), High Recharge (₹200-₹220)	Low Churn Risk (Satisfied / High ARPU)
Disengaged / Low Usage	U2, U4	Low Calls (80-100 min), Low Data (1-2 GB), Low Recharge (₹90-₹100)	High Churn Risk (Dormant / Under-utilized)

2. Root Cause Analysis for Churn Risk (U2, U4)

- Under-utilization / Inadequate Plan Fit: Users U2 and U4 consume minimal data and talktime, indicating they may be secondary SIM holders or finding current tariff plans cost-ineffective.
- Dissatisfaction / Competitor Migration: A sharp drop in data consumption often signals network coverage dissatisfaction or migration to competitor fiber/Wi-Fi.

3. Targeted Customer Retention Strategies

- For High Churn-Risk Users (U2, U4):
 1. Micro-Recharge & Bite-Sized Packs: Introduce affordable entry-level validity extension packs (e.g., ₹49/month with light data) to keep SIM active.
 2. Proactive Re-engagement Offers: Automated SMS with bonus 1GB data trial or weekend OTT bundles to stimulate data adoption.
 3. Proactive Customer Service Outreach: CSAT survey to detect network/call-drop grievances before formal port-out requests.
- For Loyal High-Usage Users (U1, U3, U5):
 1. Value Upselling: Transition to annual postpaid plans with family sharing and premium streaming subscriptions.
 2. Priority Network Experience: Offer 5G unlimited upgrades to ensure long-term retention.

Question 4: E-commerce User Clustering & Recommendation Systems

Dataset: E-commerce Behavior

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

1. Proposed Clustering Technique: K-Means / Fuzzy C-Means Clustering

Using K-Means Clustering on normalized engagement features (Views, Clicks, Purchases), the customer base cleanly segments into two behavioral groups:

Cluster Segment	Users	Behavioral Metrics	Conversion Rate (Purchases/Clicks)
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High-Intent Power Buyers	U1, U3, U5	Views: 50–60 Clicks: 20–25 Purchases: 5–6	High Conversion (~24% - 25%)
Casual / Exploratory Browsers	U2, U4	Views: 20–30 Clicks: 8–10 Purchases: 1–2	Low Conversion (~12.5% - 20%)

2. Enhancing Recommendation Systems through Clustering

Clustering powers next-generation Collaborative and Content-Based Filtering recommendation pipelines:

- Cluster-Based Collaborative Filtering (Scalability & Cold Start Resolution):
 - Instead of computing pairwise similarity across millions of users, candidate generation is constrained to the user's immediate cluster, reducing algorithmic complexity from $O(N^2)$ to $O(K \cdot N/K)$.
 - When a new user matches the profile of Cluster 1 (e.g., high views & clicks), the system immediately recommends trending products favored by U1, U3, and U5.
- Tailored Recommendation Strategies by Segment:
 - For High-Intent Segment (U1, U3, U5): Deploy personalized cross-selling, post-purchase complementary accessories, premium product upgrades, and 'Frequently Bought Together' bundles.
 - For Casual Browser Segment (U2, U4): Deploy high-urgency / conversion-boosting recommendations like 'Best Sellers', customer review highlights, flash discounts, and cart-abandonment price-drop alerts.

Question 5: Traffic Congestion Zone Identification & Traffic Management

Dataset: Traffic Data

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

1. Proposed Clustering Solution: Gaussian Mixture Models (GMM) / K-Means

To categorize urban traffic behavior, Multi-dimensional K-Means / GMM is applied on standardized metrics (Vehicle Count, Speed, Delay). The dataset partitions into two clear zones:

Congestion Zone	Locations	Traffic Characteristics	Operational Status
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Zone A: Free-Flow / Normal Corridor	L1, L3, L5	Count: 200–220 Speed: 28–32 km/h Delay: 9–12 min	Optimal Flow (Green Zone)
Zone B: High Congestion / Bottleneck	L2, L4	Count: 500–600 Speed: 10–15 km/h Delay: 25–30 min	Severe Congestion (Red Zone)

2. Analysis of Traffic Bottlenecks (L2, L4)

- High Vehicle Volume + Severe Speed Reduction: L4 and L2 experience over 2.5x the vehicle density of normal zones, causing speeds to drop to 10–15 km/h and delays to spike to 25–30 minutes.
- Gridlock Risk: The inverse non-linear correlation between vehicle density and speed signals capacity breakdown at junctions L2 and L4.

3. Actionable Traffic Management Interventions

- Dynamic Signal Control: Integrate smart adaptive traffic lights (e.g., SCATS/SCOOT) that dynamically increase green signal timing for corridors L2 and L4.
- Dynamic Route Guidance: Push live rerouting notifications to GPS navigation apps (Google Maps) to divert approaching traffic to free-flow corridors (L1, L3, L5).
- Resource & Enforcement Deployment: Station traffic wardens and rapid clearance tow vehicles at bottleneck zones during peak hours to clear stalled vehicles.
- Long-Term Infrastructure Planning: Use cluster density histories to justify road widening, flyover construction, or dedicated public transit lanes for L2/L4.